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1  cd "F:\ECON 408\Practice do file"
2
3
4  clear all
5  set seed 1000
6
7  use "HOMEWORK 3.dta" , clear
8
9  rename t time
10 tsset time
11
12
13  *                               Identification
14  ****
15
16  *(1) Time Series plot of variable y
17  //-----
18
19
20  set scheme white_tableau
21
22  tsline y, ///
23      lcolor(navy) lwidth(medthick) ///
24      title("{bf:Time Series Plot of y}", size(large)) ///
25      xtitle("Time Period", size(medlarge)) ///
26      ytitle("Values", size(medlarge)) ///
27      ylabel(, labsize(medium)) ///
28      tlabel(, labsize(medium)) ///
29      graphregion(color(white)) xsize(8) ysize(4) ///
30      name(tsplot_hw3, replace)
31
32  graph export tsplot_hw3.png, name(tsplot_hw3) replace
33
34
35  *(2) Stationarity Test
36  //-----
37
38
39  varsoc y , maxlag(12)
40  dfuller y, lags(4)
41
42
43
44  *(3) ACF and PACF
45  //-----
46
47
48  set scheme s2color
49
50  ac y, name(acf_plot,replace) lags(20)
51  pac y, name(pacf_plot, replace) lags(20)
52  graph combine acf_plot pacf_plot, name(ACF_and_PACF_hw3, replace) ///
53      title("ACF and PACF of y") xsize(8) ysize(4)
54
55  graph export ACF_and_PACF_hw3.png, name(ACF_and_PACF_hw3) replace
56
57  * Assumed models: ARMA(2,3) , ARMA(3,2), ARMA(3,3), ARMA(3,4), ARMA(4,3)
58
59  qui arima y, ar(1/2) ma(1/3)
60  estat ic
61
62  qui arima y, ar(1/3) ma(1/2)
63  estat ic

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64
65 qui arima y, ar(1/3) ma(1/3)
66 estat ic
67
68 qui arima y, ar(1/3) ma(1/4)
69 estat ic
70
71 qui arima y, ar(1/4) ma(1/3)
72 estat ic
73
74 * ARMA(2,3) has the lowest AIC and BIC
75
76
77
78 *
79 *****
80
81 arima y, ar(1/2) ma(1/3)
82 estat ic
83 estat aroots, name(arootshw3,replace)
84 graph export arootshw3.png,name(arootshw3) replace
85
86
87 *
88 *****
89
90
91 predict res , residuals
92
93 ac res, name(res_acf, replace) lags(20)
94 pac res , name(res_pacf, replace) lags(20)
95 graph combine res_acf res_pacf , xsize(8) ysize(4) title("ACF and PACF of
96 Residual") ///
97 name(res_acf_pacf_hw3, replace)
98 graph export res_acf_pacf_hw3.png, name(res_acf_pacf_hw3) replace
99
100 wntestq res
101
102
103 *
104 *****
105
106
107 keep if time <= 100
108 qui arima y, ar(1/2) ma(1/3)
109 tsappend, add(10)
110
111 capture drop y_forecast y_mse y_se ci_lower ci_upper
112 predict y_forecast, y dynamic(101)
113 predict y_mse, mse dynamic(101)
114
115 gen y_se = sqrt(y_mse)
116 gen ci_lower = y_forecast - 1.96 * y_se
117 gen ci_upper = y_forecast + 1.96 * y_se
118
119 qui replace y_forecast = y if time == 100
120 qui replace ci_lower = y if time == 100
121 qui replace ci_upper = y if time == 100
122
123 list time y_forecast ci_lower ci_upper if time >=99 & time <= 103
124
125

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126 set scheme white_tableau
127
128 twoway (rarea ci_lower ci_upper time if time >= 100, color("gs12") lwidth(
medthick)) ///
129 (line y time if time <= 100 & time >= 1, lcolor("33 37 41") lwidth(medthin
)) ///
130 (line y_forecast time if time >= 100, lcolor("255 68 69") lpattern(dash)
lwidth(medthick)), ///
131 xline(100, lcolor(gs10) lwidth(medthick) lpattern(shortdash)) ///
132 graphregion(color(white)) ///
133 title("{bf:ARMA(2,3): Historical Data vs. Dynamic Forecast}") ///
134 xtitle("Time Period") ///
135 ytitle("Value") ///
136 legend(order(2 "Actual Data" 1 "95% CI" 3 "Forecast") pos(11) ring(0) ///
cols(1)) ///
137 ylabel(, format(%9.0f) nogrid) ///
138 xlabel(0(25)100 110, labsize(small)) ///
139 xsize(6.5) ysize(4) name(forecast_plot,replace)
141
142 graph export forecast_plot.png, name(forecast_plot) replace
143
144
145
```